

$$f(x) = \arctan(x) + \frac{2x}{1+x^2}$$

D: $\arctan(x)$ ha come dominio $(-\infty, +\infty)$ e $1+x^2 > 0 \quad \forall x \in \mathbb{R}$ allora $D: (-\infty, +\infty)$

• Limiti

$$\lim_{x \rightarrow -\infty} \overbrace{\arctan(x)}^{-\frac{\pi}{2}} + \underbrace{\frac{2x}{1+x^2}}_{\rightarrow 0} = -\frac{\pi}{2}$$

$$\lim_{x \rightarrow +\infty} \arctan(x) + \frac{2x}{1+x^2} = \frac{\pi}{2}$$

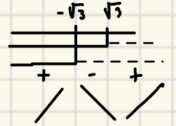
• Derivata I

$$f'(x) = \frac{1}{1+x^2} + \frac{2(1+x^2) - 2x(2x)}{(1+x^2)^2} = \frac{1}{1+x^2} + \frac{2-2x^2}{(1+x^2)^2} = \frac{1+x^2+2-2x^2}{(1+x^2)^2} = \frac{3-x^2}{(1+x^2)^2}$$

$$f'(x) > 0$$

$$\frac{3-x^2}{(1+x^2)^2} \Rightarrow 3-x^2 > 0 \Rightarrow 3 > x^2 \Rightarrow \sqrt{3} > x \Rightarrow$$

Denominatore sempre positivo

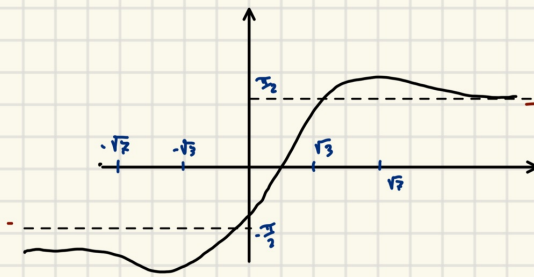
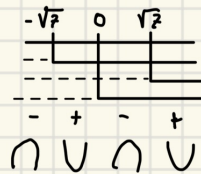


Derivata II

$$\left(\frac{3-x^2}{(1+x^2)^2}\right)' = \frac{-2x(1+x^2)^2 - (3-x^2)2(1+x^2)2x}{(1+x^2)^4} = \frac{-2x(1+x^2) - (3-x^2)4x}{(1+x^2)^3} = \frac{-2x-2x^3-12x+4x^3}{(1+x^2)^3} = \frac{2x^3-14x}{(1+x^2)^3}$$

$$\frac{2x^3-14x}{(1+x^2)^3} > 0 \Rightarrow x(2x^2-14) > 0 \Rightarrow x > 0$$

$$x^2 > 7 \Rightarrow x > \pm\sqrt{7}$$



11.2

$$f(x) = \sqrt{\cos(2x+x^2) + \frac{x \sin(x)}{1+x^2}} =$$

$$\sqrt{1-2x^2-x^3-\frac{x^4}{2}+\frac{2x^4}{3}+\frac{x^2-\frac{x^4}{6}+o(x^4)}{1+x^2}} =$$

$$\sqrt{1-2x^2-x^3+\frac{x^4}{6}+x^2-\frac{x^4}{6}(1-x^2+x^4)+o(x^4)} =$$

$$\sqrt{1-2x^2-x^3+\frac{x^4}{6}+\left(x^2-\frac{x^4}{6}\right)(1-x^2+x^4)} =$$

$$\sqrt{1-2x^2-x^3+\frac{x^4}{6}+\frac{x^4}{6}-\frac{x^6}{6}-x^4+o(x^4)} = \sqrt{1-x^2-x^3-x^4} = 1 + \frac{1}{2}(-x^2-x^3-x^4) + \frac{1}{2}\left(\frac{1}{2} \cdot \frac{1}{2}\right)(-x^2-x^3-x^4)^2 =$$

$$1 - \frac{x^2}{2} - \frac{x^3}{2} - \frac{x^4}{2} + \frac{1}{8}(-x^4) =$$

$$1 - \frac{x^2}{2} - x^3 - \frac{5x^4}{8}$$

$$\cos(2x+x^2) = 1 - \frac{(2x+x^2)^2}{2} + \frac{(2x+x^2)^4}{4!} = 1 - \frac{4x^2+2x^3+x^4}{2} + \frac{2x^4}{3}$$

$$\sin(x) = x - \frac{x^3}{3!} + o(x^3)$$

$$(1+x^2)^{-1} = 1 - x^2 + \frac{1(1-1)}{2}x^4 + o(x^4)$$

N. 3

$$f(x) = e^{-2x} \log(3 + e^{x+|x|})$$

a) Determinare x per cui f è strettamente decrescente da $(x, +\infty)$

Sappiamo che la monotonia di $f(x)$ è determinata dalla derivata f' , per cui:

$$f'(x) = \begin{cases} x < 0 & ((e^{-2x} \log(3 + 1))' = [e^{-2x} \cdot (-2) + \log(4)] + e^{-2x} = e^{-2x}(-2 + 2\log(2) + 1) \\ x > 0 & ((e^{-2x} \log(3 + e^{2x}))' = [e^{-2x} \cdot (2) \cdot \log(3 + e^{2x}) + e^{-2x} \cdot \frac{1}{3 + e^{2x}} \cdot e^{2x} \cdot 2x] = e^{-2x}[-2\log(3 + e^{2x}) + \frac{2xe^{2x}}{3 + e^{2x}}] \end{cases}$$

Poiché viene chiesto un $x > 0$ per $(x, +\infty)$ consideriamo solo l'intervallo $x > 0$

$$e^{-2x} \left[-2\log(3 + e^{2x}) + \frac{2xe^{2x}}{3 + e^{2x}} \right] > 0$$

$e^{-2x} > 0 \quad \forall x \in \mathbb{R}$ poiché esso è sempre positivo basta determinare il segno del secondo fattore per $x \rightarrow +\infty$

$$\lim_{x \rightarrow +\infty} -2\log(3 + e^{2x}) + \frac{2xe^{2x}}{3 + e^{2x}} \sim -4x \log(e) + \frac{e^{2x}}{e^{2x}} = -4(+\infty)(1) = -\infty$$

b)

$$\int_{-\log(2)}^{+\infty} e^{-2x} \log(3 + e^{x+|x|}) dx$$

Abbiamo:

$$F(x) = \int_{-\log(2)}^0 e^{-2x} \log(4) dx + \int_0^{+\infty} e^{-2x} \log(3 + e^{2x}) dx = \log(4) \int_{-\log(2)}^0 e^{-2x} dx + \int_0^{+\infty} e^{-2x} \log(3 + e^{2x}) dx =$$

$$\log(4) \left[\frac{e^{-2x}}{-2} \right]_{-\log(2)}^0 + \int_0^{+\infty} e^{-2x} \log(3 + e^{2x}) dx =$$

$$\log(4) \left(-\frac{1}{2} + \frac{1}{2} \right) + \frac{1}{2} \int_0^1 \frac{\log(3+t)}{t} dt$$

$$\log(4) \cdot \frac{3}{2} - \frac{1}{2} \left[-\frac{\log(3+t)}{t} \right]_1^{+\infty} + \frac{1}{2} \int_1^{+\infty} \frac{1}{3+t} \cdot \frac{1}{t} dt$$

$$\frac{3}{2} \ln(4) - \frac{1}{2} \left[\frac{\log(3+t)}{t} \right]_1^{+\infty} - \frac{1}{6} \int_1^{+\infty} \frac{1}{t} dt + \frac{1}{3} \int_1^{+\infty} \frac{1}{t+3} dt =$$

$$\frac{3}{2} \log(4) - \frac{1}{2} (-\log(4)) + \frac{1}{6} \ln(3+t) \Big|_1^{+\infty} - \frac{1}{6} \ln(t) \Big|_1^{+\infty}$$

$$\frac{3}{2} \log(4) + \frac{4 \log(2)}{3} = 3 \log(2) + \frac{4 \log(2)}{3} = \frac{9 \log(2) + 4 \log(2)}{3} = \frac{13 \log(2)}{3}$$

$$\boxed{\begin{aligned} \frac{A}{t} + \frac{B}{3+t} \\ \frac{3A + At + Bt}{t(3+t)} \quad \begin{cases} B = -\frac{1}{3} \\ A = \frac{1}{3} \end{cases} \end{aligned}}$$

4)

$$\overline{z} (|z|^2 - 5) = 6$$

$$(x - iy) ((\sqrt{x^2 + y^2})^2 - 5) = 6$$

$$(x - iy)^2 (x^2 + y^2 - 5) = 6$$

$$(x^2 - 2xy \cdot y^2) (x^2 + y^2 - 5) = 6$$

$$\begin{cases} (x^2 - y^2) (x^2 + y^2 - 5) = 6 \\ (-2xy) (x^2 + y^2 - 5) = 6 \end{cases} \quad \begin{cases} -y^2 (y^2 - 5) = 6 \\ x = 0 \end{cases} \quad \cup \quad \begin{cases} x^2 (x^2 - 5) = 6 \\ y = 0 \end{cases} \quad \cup \quad \begin{cases} (x^2 - y^2) 0 = 6 \\ x^2 + y^2 = 5 \end{cases}$$

$$\begin{cases} y^4 - 5y^2 + 6 \\ x = 0 \end{cases} \quad \cup \quad \begin{cases} x^4 - 5x^2 + 6 \\ y = 0 \end{cases} \quad \cup \quad \emptyset$$

$$\begin{cases} y \in \{\pm\sqrt{2}, \pm\sqrt{3}\} \\ x = 0 \end{cases} \quad \cup \quad \begin{cases} x \in \{\pm\sqrt{6}\} \\ y = 0 \end{cases}$$

$$\pm\sqrt{2}i, \pm\sqrt{3}i, \pm\sqrt{6}$$